

ITA0609 - MACHINE LEARNING

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1..A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

Given:

Objective Function: $f(x) = x^2$

Initial Population: 2, 4, 6, 8

Step 1: Calculate Fitness Values

The fitness of each individual is calculated using the function $f(x) = x^2$.

For $x = 2$:

$$\text{Fitness} = 2^2 = 4$$

For $x = 4$:

$$\text{Fitness} = 4^2 = 16$$

For $x = 6$:

$$\text{Fitness} = 6^2 = 36$$

For $x = 8$:

$$\text{Fitness} = 8^2 = 64$$

Therefore, the fitness values are:

- Individual 2 $\rightarrow$ Fitness = 4
- Individual 4 $\rightarrow$ Fitness = 16
- Individual 6 $\rightarrow$ Fitness = 36
- Individual 8 $\rightarrow$ Fitness = 64

Step 2: Selection

The individuals with the highest fitness values are selected as parents.

Selected Parents:

- 8 (Fitness = 64)
- 6 (Fitness = 36)

Step 3: Binary Representation

8 = 1000

6 = 0110

Step 4: Crossover

The least significant bit (LSB) of both parents is exchanged.

Parent 1 = 1000

Parent 2 = 0110

Since both least significant bits are 0, exchanging them does not change the chromosomes.

Therefore, the offspring are:

Offspring 1 = 1000 (Decimal 8)

Offspring 2 = 0110 (Decimal 6)

Step 5: Role of Mutation

Mutation is a genetic operator that randomly changes one or more bits in a chromosome after crossover.

Example:

Before Mutation:
1000

After Mutation:
1001

Mutation plays an important role in Genetic Algorithms because it:

- Introduces new genetic information into the population.
- Maintains diversity among individuals.
- Prevents premature convergence to local optimum solutions.
- Helps the algorithm explore a larger search space.
- Increases the probability of finding the global optimum solution.

Conclusion

The calculated fitness values are 4, 16, 36, and 64. The best individuals selected are 8 and 6. After performing crossover by exchanging the least significant bits, the offspring remain unchanged because both bits are identical. Mutation improves the diversity of the population and helps the Genetic Algorithm produce better solutions.

2. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is -0.4 . For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.

Given:

Weight 1 (w_1) = 0.5

Weight 2 (w_2) = 0.3

Bias (b) = -0.4

Advertisement Exposure (x_1) = 2

Income Level (x_2) = 1

Step 1: Calculate Net Input

Formula:

$$\text{Net Input} = (w_1 \times x_1) + (w_2 \times x_2) + b$$

Substituting the given values:

$$\text{Net Input} = (0.5 \times 2) + (0.3 \times 1) + (-0.4)$$

$$\text{Net Input} = 1.0 + 0.3 - 0.4$$

$$\text{Net Input} = \mathbf{0.9}$$

Step 2: Apply Step Activation Function

The step activation function is defined as:

- If Net Input ≥ 0 , Output = 1
- If Net Input < 0 , Output = 0

Since the net input is **0.9**, which is greater than zero,

$$\text{Output} = 1$$

Step 3: Decision

The output of the perceptron is **1**, indicating that the customer **will purchase the product**.

Conclusion

The net input to the perceptron is **0.9**. After applying the step activation function, the output is **1**. Therefore, the perceptron predicts that the customer **will purchase the product**.